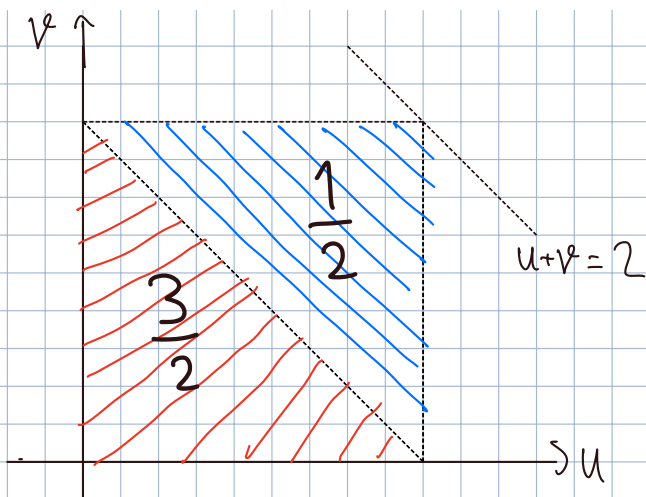


ECE 461 - HW02

Boran Aybak Kiliç

1)

$$f_{X,Y}(u,v) = \begin{cases} \frac{3}{2}, & 0 \leq u < 1, 0 \leq v < 1, 0 \leq u+v < 1, \\ \frac{1}{2}, & 0 \leq u < 1, 0 \leq v < 1, 1 \leq u+v < 2, \end{cases}$$



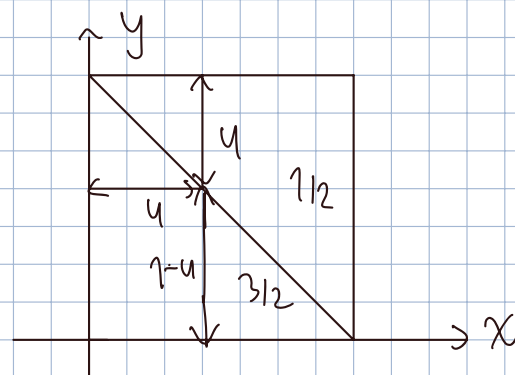
We can also define

1st region: $u \in [0,1], v \in [0, 1-u]$

2nd region: $u \in [0,1], v \in [1-u, 1]$

$$1.a) \quad f_X(u) = \int_{-\infty}^{\infty} f_{X,Y}(u,v) dv$$

→ If $u \notin [0,1]$, then $f_X(u) = 0$ clearly
since $f_{X,Y}(u,v) = 0$.



For $u \in [0,1]$

$$f_X(u) = \int_0^{1-u} \frac{3}{2} dy + \int_{1-u}^1 \frac{1}{2} dy = \frac{3}{2} (1-u) + \frac{1}{2} u = \frac{3}{2} - u$$

just a free parameter
in u's direction

$$f_X(u) = \begin{cases} \frac{3}{2} - u, & 0 \leq u \leq 1 \\ 0, & \text{else} \end{cases}$$

$$1.b) \quad f_{Y|X}(v|u) = \frac{f_{X,Y}(u,v)}{f_X(u)}$$

⚠ Note that $f_X(u) = 0$ for $u \notin [0,1]$

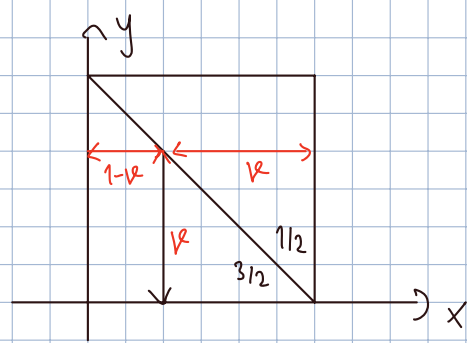
$$= \begin{cases} \frac{3}{3-2u} ; & u,v \in [0,1], v \in [0, 1-u] \\ \frac{1}{3-2u} ; & u,v \in [0,1], v \in [1-u, 1] \\ 0 ; & u \in [0,1], v \notin [0,1] \\ \text{undefined} ; & u \notin [0,1] \end{cases}$$

$$1.c) f_Y(v) = \int_{-\infty}^{\infty} f_{X,Y}(u,v) du$$

→ For $v \notin [0,1]$, $f_Y(v) = 0$

For $v \in [0,1]$:

$$f_Y(v) = \int_0^{1-v} \frac{3}{2} dx + \int_{1-v}^1 \frac{1}{2} dx = \frac{3}{2} - v$$



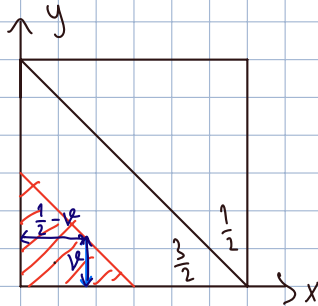
$$f_Y(v) = \begin{cases} \frac{3-2v}{2} & , 0 \leq v \leq 1 \\ 0 & , \text{else} \end{cases}$$

$$1.d) f_{X|Y}(u|v) = \frac{f_{X,Y}(u,v)}{f_Y(v)}$$

$$f_{X|Y}(u|v) = \begin{cases} \frac{3}{3-2u} & , u,v \in [0,1], u \in [0, 1-v] \\ \frac{1}{3-2u} & , u,v \in [0,1], u \in [1-v, 1] \\ 0 & , v \in [0,1], u \notin [0,1] \\ \text{undefined} & , v \notin [0,1] \end{cases}$$

* Same as b)
with u & v
interchanged
since
pdf is symmetric
w.r.t. u & v

$$1.e) P\{X+Y \leq \frac{1}{2}\} = \iint_{X+Y \leq \frac{1}{2}} f_{X,Y}(u,v) du dv = \int_0^{1/2} \int_0^{\frac{1}{2}-v} \frac{3}{2} du dv$$



$$= \int_0^{1/2} \frac{3}{2} \left(\frac{1}{2}-v\right) dv = \left[\frac{3}{4} v - \frac{3}{4} v^2 \right]_0^{1/2} = \frac{3}{8} - \frac{3}{16} = \boxed{\frac{3}{16}}$$

$$2) \quad f_{y|x}(y|x) = \frac{(x+\lambda)^y}{y!} e^{-(x+\lambda)}, \quad x \geq 0, \quad y \in \mathbb{Z}^+$$

with $0 \leq x \leq A$ and $E[X] = \epsilon$

$$2.a) \quad P_X(x) = \begin{cases} \epsilon/A & , \quad x = A \\ 1 - \epsilon/A & , \quad x = 0 \end{cases}$$

$$E[X] = \sum_{x \in \{0, A\}} x P_X(x) = \frac{\epsilon}{A} \cdot A + \left(1 - \frac{\epsilon}{A}\right) \cdot 0 = \boxed{\epsilon}$$

2.b) Let H_0 be the hypothesis that $x=0$
 H_1 " " " $x=A$

We choose H_0 if $\frac{f_{y|x}(y|x=0)}{f_{y|x}(y|x=A)} \geq \frac{P(x=A)}{P(x=0)}$, H_1 otherwise

$$f_{y|x}(y|x=0) = \frac{\lambda^y}{y!} e^{-\lambda}, \quad f_{y|x}(y|x=A) = \frac{(\lambda+A)^y}{y!} e^{-(\lambda+A)}, \quad P(X=A) = \frac{\epsilon}{A}, \quad P(X=0) = 1 - \frac{\epsilon}{A}$$

$$\frac{\lambda^y e^{-\lambda}}{(\lambda+A)^y e^{-\lambda} e^{-A}} = \left(\frac{\lambda}{\lambda+A}\right)^y e^A \geq \frac{\epsilon}{A-\epsilon}$$

Taking logarithm of both sides: $y \ln\left(\frac{\lambda}{\lambda+A}\right) + A \geq \ln\left(\frac{\epsilon}{A-\epsilon}\right)$

$$y \leq \frac{-A + \ln(\epsilon/(A-\epsilon))}{\ln\left(\frac{\lambda}{\lambda+A}\right)} \Rightarrow \begin{array}{l} \text{greater becomes} \\ \text{smaller since sign} \\ \text{of } \ln\left(\frac{\lambda}{\lambda+A}\right) \\ \text{is negative.} \end{array}$$

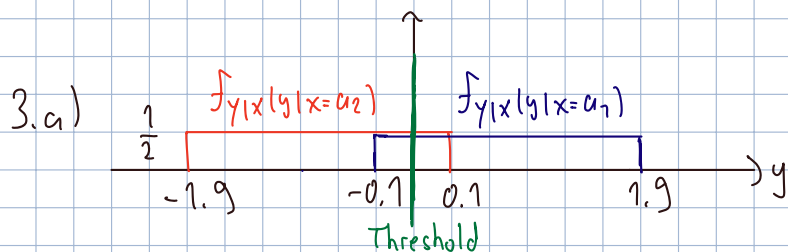
MAP rule is:

We decide $x=0$ when $y \leq \frac{-A + \ln(\epsilon/(A-\epsilon))}{\ln\left(\frac{\lambda}{\lambda+A}\right)}$, we decide A otherwise

$$3) Y = X + W \quad f_{Y|X}(y|X=a_1) = \begin{cases} 1/2, & -0.1 \leq y \leq 1.9 \\ 0, & \text{else} \end{cases}$$

$$a_1 = +1, a_2 = -1$$

$$f_{Y|X}(y|X=a_2) = \begin{cases} 1/2, & -1.9 \leq y \leq 0.1 \\ 0, & \text{else} \end{cases}$$



→ In the case of equally likely symbols optimal decision rule is ML and it simplifies to "Nearest Neighbor Rule"

$$\text{Choose } \begin{cases} a_1 & \text{if } Y \geq 0 \\ a_2 & \text{if } Y < 0 \end{cases}$$

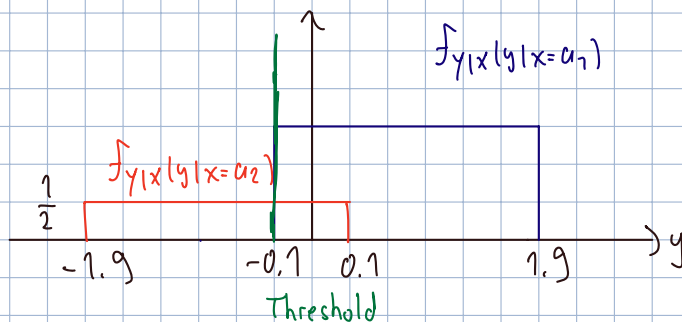
decide a_1 $\uparrow$ symbol is a_2 ($X=a_2$)

$$P_e = P(Y_1|X_2)P(X_2) + P(Y_2|X_1)P(X_1)$$

$$= \underbrace{P(Y \geq 0|X_2)}_{\int_0^{0.1} f_{Y|X}(y|X=a_2) dy = 0.05} \frac{1}{2} + \underbrace{P(Y < 0|X_1)}_{\int_{-0.1}^0 f_{Y|X}(y|X=a_1) dy = 0.05} \frac{1}{2} = \boxed{0.05}$$

3.b) MAP rule is

$$\text{Choose } \begin{cases} a_1 & \text{if } Y \geq -0.1 \\ a_2 & \text{if } Y < -0.1 \end{cases}$$



$$P_e = \underbrace{P(Y_1|X_2)P(X_2)}_{\int_{-0.1}^{0.1} f_{Y|X}(y|X=a_2) dy = 0.1} + \underbrace{P(Y_2|X_1)P(X_1)}_{\text{zero}} = 0.1 \times \frac{1}{4} = \boxed{0.025}$$

→ since a_1 has higher prob. we'll always choose it for $Y \geq -0.1$

3.c) For $P(X=a_1) = \alpha$, $P(X=a_2) = 1-\alpha$

↳ MAP rule will change depending on whichever α , $1-\alpha$ is bigger.

First suppose $\alpha > 0.5$

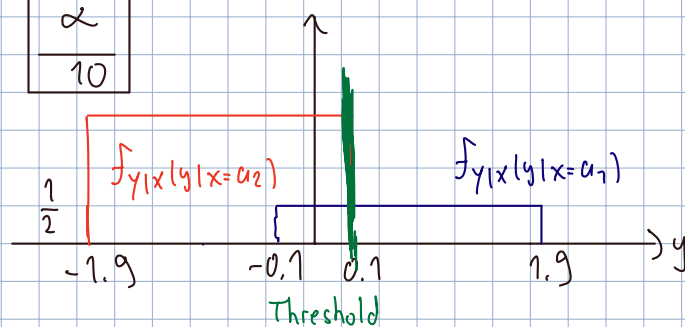
MAP rule is choose $\begin{cases} X=a_1 & \text{if } Y \geq -0.1 \\ X=a_2 & \text{if } Y < -0.1 \end{cases}$

$$P_e = \underbrace{P(Y_1|X_2)P(X_2)}_{\int_{-0.1}^{0.1} f_{Y|X}(y|x=a_2) dy = 0.1} + \underbrace{P(Y_2|X_1)P(X_1)}_{\text{zero similar to 2.b}} = 0.1 \times 1-\alpha = \boxed{\frac{1-\alpha}{10}}$$

Suppose $\alpha < 0.5$

MAP rule is choose $\begin{cases} X=a_1 & \text{if } Y \geq +0.1 \\ X=a_2 & \text{if } Y < +0.1 \end{cases}$

$$P_e = \underbrace{P(Y_1|X_2)P(X_2)}_{\text{zero}} + \underbrace{P(Y_2|X_1)P(X_1)}_{\int_{-0.1}^{0.1} \frac{1}{2} dy = 0.1} = \boxed{\frac{\alpha}{10}}$$



3.d) Since we are using the same decision rules, threshold for detection will remain the same.

$\alpha > \frac{1}{2}$ case

$$P_e = \underbrace{P(Y_1|X_2)P(X_2)}_{0.1 \text{ again}} + \underbrace{P(Y_2|X_1)P(X_1)}_{1-\beta \text{ now}} = \boxed{\frac{1-\beta}{10}}$$

$\alpha < \frac{1}{2}$ case

$$P_e = \underbrace{P(Y_1|X_2)P(X_2)}_{\text{zero again}} + \underbrace{P(Y_2|X_1)P(X_1)}_{0.1 \text{ again } \beta \text{ now}} = \boxed{\frac{\beta}{10}}$$

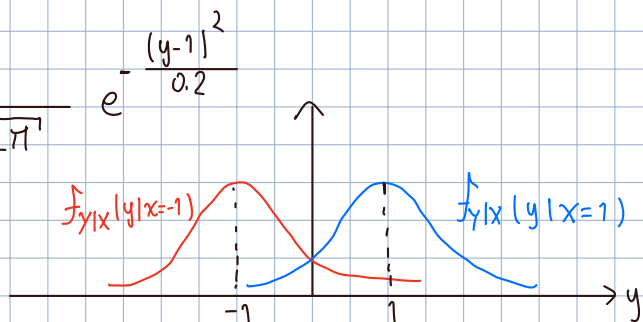
$$4) Y = X + W \quad \text{where} \quad W \sim \mathcal{N}(0, \overset{\rightarrow \sigma^2}{0.1})$$

Let X_1, X_2 be the events that $X = a_1 = +1$ and $X = a_2 = -1$, respectively.

4.a) For equally likely symbols, optimal decision rule is ML (maximum likelihood)
If a constant $X = a_i$ term is added to the Gaussian noise, received signal will be a Gaussian with mean a_i , s.t.d. 0.1

$$Y|X=a_1 \sim \mathcal{N}(1, 0.1) \Rightarrow f_{Y|X}(y|x=1) = \frac{1}{\sqrt{0.2\pi}} e^{-\frac{(y-1)^2}{0.2}}$$

$$Y|X=a_2 \sim \mathcal{N}(-1, 0.1) \Rightarrow f_{Y|X}(y|x=-1) = \frac{1}{\sqrt{0.2\pi}} e^{-\frac{(y+1)^2}{0.2}}$$



We choose X_1 if $\frac{f_{Y|X}(y|x=1)}{f_{Y|X}(y|x=-1)} = \exp\left[-\frac{(y-1)^2 - (y+1)^2}{0.2}\right] = \exp[20y] \geq 1 \Rightarrow y \geq 0$

ML rule is; we choose	$\begin{cases} a_1 = +1 & \text{if } Y \geq 0 \\ a_2 = -1 & \text{if } Y < 0 \end{cases}$
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$$4.b) P_e = \underbrace{P(Y_2|X_1)}_{P(\text{error}|X_1)} P(X_1) + \underbrace{P(Y_1|X_2)}_{P(\text{error}|X_2)} P(X_2)$$

$$P(Y_2|X_1) = P(Y < 0|X_1) = \Phi\left(\frac{0-1}{\sqrt{0.1}}\right)$$

$$P(Y_1|X_2) = P(Y > 0|X_2) = Q\left(\frac{0+1}{\sqrt{0.1}}\right)$$

both equal to $\Phi(-\sqrt{10} = -3.162) = 7.83 \times 10^{-4}$

4.c) Optimum decision rule is MAP with condition for choosing $X=a_1$ is

$$\frac{f_{Y|X}(y|X=a_1)}{f_{Y|X}(y|X=a_2)} = \exp[20y] \geq \frac{P(X_2)}{P(X_1)} = \frac{1}{3} \Rightarrow y = -\frac{1}{20} \ln 3 = -0.0549$$

MAP rule is we decide $\begin{cases} X=a_1, & Y \geq -0.0549 \\ X=a_2, & Y < -0.0549 \end{cases} \Rightarrow$ Since $f_{Y|X}(y|X=+1)$ has higher weight, decision thres. shifted to the left

4.d)
$$\begin{aligned} P_e &= P(\text{error}|X_1)P(X_1) + P(\text{error}|X_2)P(X_2) \\ &= P(Y < -0.0549|X_1) \times \frac{3}{4} + P(Y > -0.0549|X_2) \times \frac{1}{4} \\ &= \Phi\left(\frac{-0.0549-1}{\sqrt{0.1}}\right) \times \frac{3}{4} + Q\left(\frac{-0.0549+1}{\sqrt{0.1}}\right) \times \frac{1}{4} = 6.686 \times 10^{-4} \end{aligned}$$

$\Phi(-3.336) = Q(3.336)$ $Q(2.989)$

↳ MATLAB's `erf` function was used to evaluate these

5)

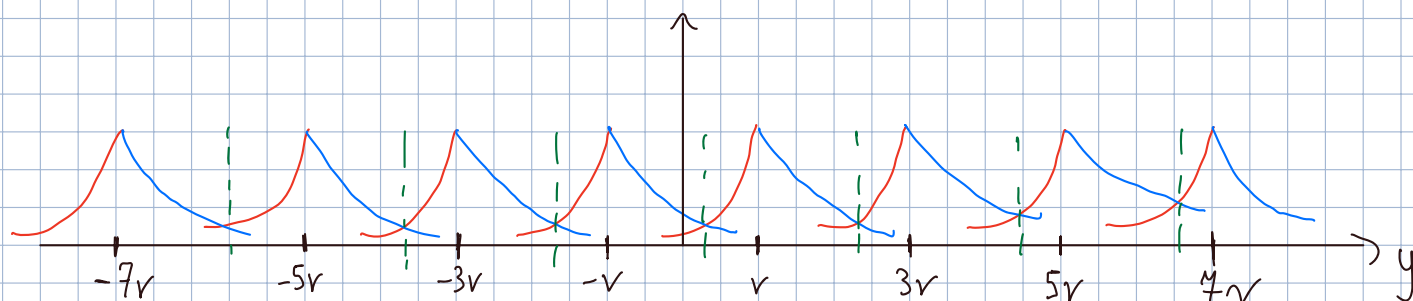
$$v_k = -7v + 2vk \quad \text{for } k = 0, 1, \dots, 7 \Rightarrow v_k = -7v, -5v, \dots, 3v, 5v, 7v$$

$$X = v_k, \quad Y = X + W \quad \text{where} \quad f_w(u) = \begin{cases} ce^{2u} & , u < 0 \\ ce^{-u} & , u \geq 0 \end{cases}$$

5.a) We require $\int_{-\infty}^{\infty} du f_w(u) = 1$

$$c \int_{-\infty}^0 du e^{2u} + c \int_0^{\infty} du e^{-u} = \frac{c}{2} + c = \frac{3c}{2} = 1 \Rightarrow \boxed{c = \frac{2}{3}}$$

5.b)



Adding a definite value X to a random variable W will only shift its pdf but shape of pdf will remain the same

$$f_{X_k}(u) = \begin{cases} \frac{2}{3} e^{2(u-v_k)} & , u < v_k \\ \frac{2}{3} e^{-(u-v_k)} & , u \geq v_k \end{cases}$$

→ For 2 adjacent symbols X_k and X_{k-1} , threshold value between them is found by setting left-side pdf of X_k to right-side pdf of X_{k-1}

$$\frac{2}{3} e^{2(u_k^* - v_k)} = \frac{2}{3} e^{-(u_k^* - v_{k-1})} \Rightarrow 2u_k^* - 2v_k = -u_k^* + v_{k-1} \Rightarrow u_k^* = \frac{2v_k + v_{k-1}}{3} //$$

↳ Note that this is only valid for $k = 1, \dots, 7$ (and not 0)

Then thresholds will be at $u_k^* = \frac{2(-7v + 2vk) + (-7v + 2v(k-1))}{3} = -7v \cdot \frac{2v}{3} + 2vk$

$$= \left(-\frac{23}{3} + 2k\right)v$$

Then ML decision boundaries becomes

$$R_k = \begin{cases} (-\infty, -\frac{17}{3}v) & , & \text{for } k=0 \\ \left[\left(-\frac{23}{3} + 2k\right)v, \left(-\frac{17}{3} + 2k\right)v\right] & , & \text{for } k=1, 2, \dots, 6 \\ \left(\frac{19}{3}v, \infty\right) & , & \text{for } k=7 \end{cases}$$

For error prob., let us first calculate P_e of intermediary symbols ($k=1, \dots, 6$)

$$\begin{aligned} P(\text{error}|v_k) &= P_r\left\{Y < \left(-\frac{23}{3} + 2k\right)v \mid X=v_k\right\} + P_r\left\{Y > \left(-\frac{17}{3} + 2k\right)v \mid X=v_k\right\} \\ &= \int_{-\infty}^{\left(-\frac{23}{3} + 2k\right)v} \frac{2}{3} e^{2(u-v_k)} du + \int_{\left(-\frac{17}{3} + 2k\right)v}^{\infty} \frac{2}{3} e^{-|u-v_k|} du \\ &= \frac{2}{3} \cdot \frac{1}{2} e^{-2v_k} e^{2\left(-\frac{23}{3} + 2k\right)v} + \frac{2}{3} e^{v_k} e^{-\left(-\frac{17}{3} + 2k\right)v} \\ &= \frac{1}{3} e^{2\left(-\frac{23}{3} + 2k + 7 - 2k\right)v} + \frac{2}{3} e^{+(-7 + 2k + \frac{17}{3} - 2k)v} = \frac{1}{3} e^{-\frac{4}{3}v} + \frac{2}{3} e^{-\frac{4}{3}v} = e^{-\frac{4}{3}v} \end{aligned}$$

For $k=0 \Rightarrow P(\text{error}|v_0) = \int_{-\frac{17}{3}v}^{\infty} \frac{2}{3} e^{-(u-v_0)} du = \frac{2}{3} e^{-\left(\frac{17}{3} + 7\right)v} = \frac{2}{3} e^{-4v/3}$

For $k=7 \Rightarrow P(\text{error}|v_7) = \int_{-\infty}^{\frac{19}{3}v} \frac{2}{3} e^{2(u-v_7)} du = \frac{1}{3} e^{-4v/3}$

$$P_e = \frac{1}{8} \sum_{k=0}^7 P(\text{error}|v_k) = \frac{1}{8} \left[6 \times e^{-4/3} + \frac{2}{3} e^{-4/3} + \frac{1}{3} e^{-4/3} \right] = \frac{7}{8} e^{-4v/3}$$

$\underbrace{\frac{1}{8}}_{P(v_k)}$

Average energy per bit is just

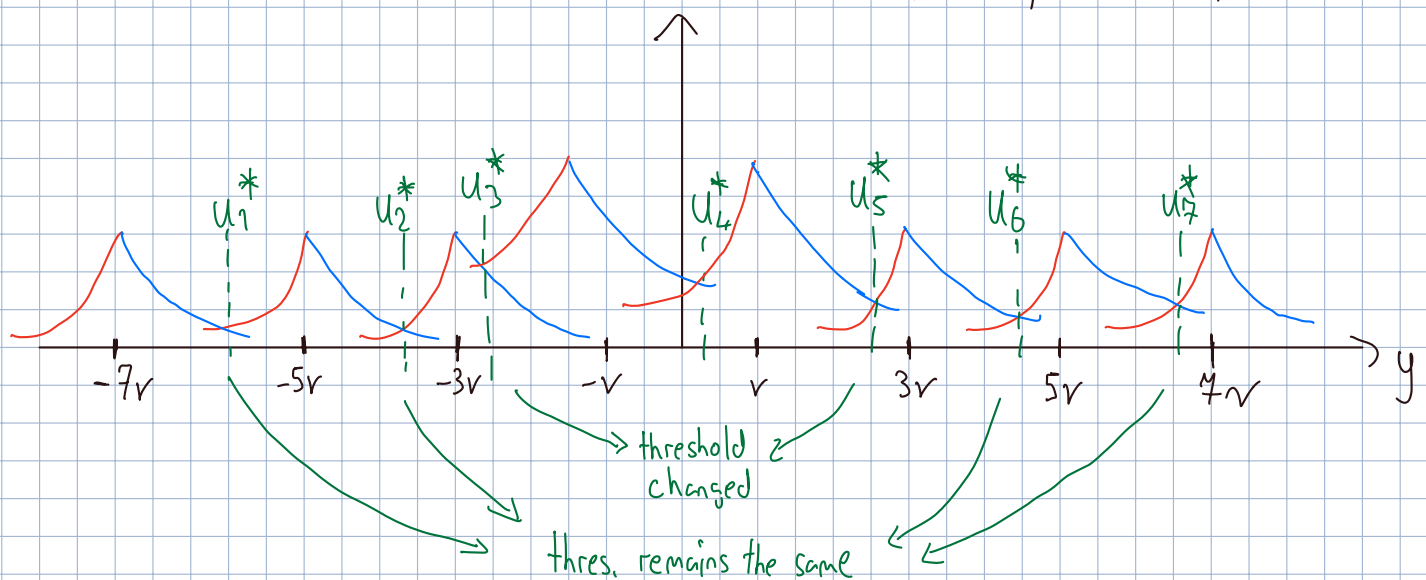
$$E_{\text{ave}} = \underbrace{\frac{1}{3}}_{\text{\# of bits}} \underbrace{\sum_{k=0}^7 (V_k)^2 P(V_k)}_{\text{total ave energy}} = \frac{1}{3} \cdot \frac{1}{8} \cdot 2 \left[(7v)^2 + (5v)^2 + (3v)^2 + (v)^2 \right] = \boxed{7v^2}$$

5.c) Minimizing energy per bit

Since symbols 001 and 100 are twice as likely, we need to assign them to lowest abs. value voltage levels $\pm v$. Rest of the symbols can assigned arbitrarily to remaining voltages.

New priori probs are :

$$P(V_k) = \begin{cases} 1/5, & k=3,4 \\ 1/10, & k=0,1,2,5,6,7 \end{cases}$$



→ Notice that only thresholds that change is u_3^* , u_5^* and possibly u_4^* . Others will remain the same.

For u_3^* :

$$\frac{1}{5} e^{2(u_3^* - v)} = \frac{1}{10} e^{-(u_3^* - 3v)} \Rightarrow 2 = e^{-3u_3^* - 5v}$$

$$\ln 2 = -3u_3^* - 5v$$

$$u_3^* = \frac{-\ln 2 - 5v}{3}$$

For u_5^* :

$$\frac{1}{5} e^{-(u_5^* - v)} = \frac{1}{10} e^{2(u_5^* - 3v)} \Rightarrow \ln 2 = 3u_5^* - 7v$$

$$u_5^* = \frac{\ln 2 + 7v}{3}$$

→ Note that for u_4^* $\frac{1}{5} c e^{-(u+v)} = \frac{1}{5} c e^{2(u-v)}$ still yields $u_4^* = \frac{v}{3}$

→ $P(\text{error} | V_k)$ remains unchanged for $k=0, 1, 6, 7$

$$P(\text{error} | V_2) = \int_{-\infty}^{-\frac{11v}{3}} c e^{2(u+3v)} dv + \int_{-\frac{\ln 2 - 5v}{3}}^{\infty} c e^{-(u+3v)} dv$$

$$= \underbrace{\frac{1}{3} e^{-4v/3}}_{\text{from previous option}} + \frac{2}{3} \underbrace{e^{\frac{\ln 2}{3}}}_{2^{1/3}} e^{-4v/3} = \frac{e^{-4v/3}}{3} (1 + 2^{4/3})$$

$$P(\text{error} | V_3) = \int_{-\infty}^{-\frac{\ln 2 - 5v}{3}} c e^{2(u+v)} dv + \int_{v/3}^{\infty} c e^{-(u+v)} dv$$

$$= \frac{1}{3} e^{-2\frac{\ln 2}{3}} e^{-4v/3} + \underbrace{\frac{2}{3} e^{-4v/3}}_{\text{from previous option}} = \frac{e^{-4v/3}}{3} (2 + 2^{-2/3})$$

$$P(\text{error} | V_4) = \int_{-\infty}^{v/3} c e^{2(u-v)} dv + \int_{\frac{\ln 2 + 7v}{3}}^{\infty} c e^{-(u-v)} dv$$

$$= \frac{1}{3} e^{-4v/3} + \frac{2}{3} e^{-4v/3} \underbrace{e^{-\frac{\ln 2}{3}}}_{2^{-1/3}} = \frac{e^{-4v/3}}{3} (1 + 2^{2/3})$$

$$P(\text{error} | V_5) = \int_{-\infty}^{\frac{\ln 2 + 7v}{3}} c e^{2(u-3v)} dv + \int_{\frac{13v}{3}}^{\infty} c e^{-(u-3v)} dv$$

$$= \frac{1}{3} e^{-4v/3} e^{2\frac{\ln 2}{3}} + \frac{2}{3} e^{-4v/3} = \frac{e^{-4v/3}}{3} (2 + 2^{2/3})$$

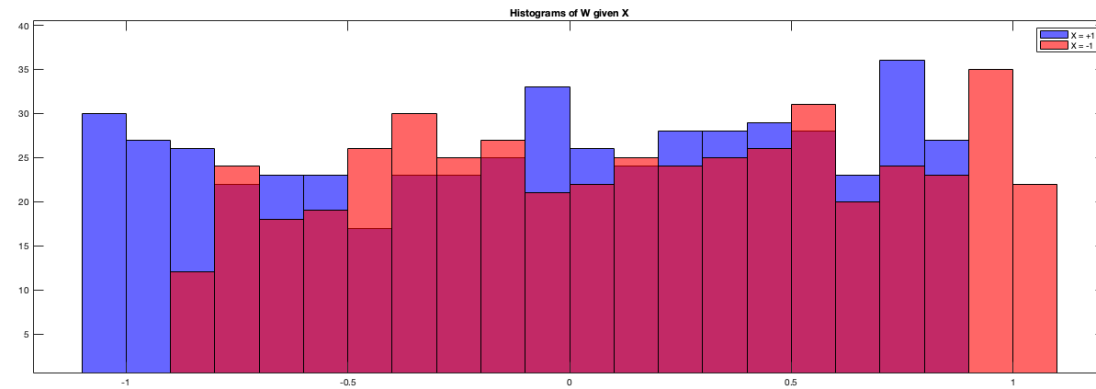
$$\begin{aligned}
 P_e &= \frac{1}{10} \times \underbrace{2 e^{-4v/3}}_{k=2,6} + \frac{1}{10} \times \underbrace{\frac{2}{3} e^{-4v/3}}_{k=0} + \frac{1}{10} \times \underbrace{\frac{1}{3} e^{-4v/3}}_{k=7} \\
 &= \frac{1}{5} \times \frac{e^{-4v/3}}{3} \left[\underbrace{2+2}_{k=3}^{-2/3} + \underbrace{1+2}_{k=4}^{2/3} \right] + \frac{1}{10} \times \frac{e^{-4v/3}}{3} \left[\underbrace{1+2}_{k=2}^{4/3} + \underbrace{2+2}_{k=5}^{2/3} \right] \\
 &= \frac{e^{-4v/3}}{10} \left[6 + \frac{2^{1/3}}{3} + \frac{2^{5/3}}{3} + \frac{2^{4/3}}{3} + \frac{2^{2/3}}{3} \right] = \frac{6 + 2^{1/3} + 2^{2/3}}{10} e^{-4v/3} \approx 0.885 e^{-4v/3}
 \end{aligned}$$

Finally, average energy per bit is:

$$E_{ave} = \frac{1}{3} \times \left\{ \frac{1}{10} \times \left[2 \times (7v)^2 + 2 \times (5v)^2 + 2 \times (3v)^2 \right] + \frac{1}{5} \times 2 \times (v)^2 \right\} = \frac{2 \times 85}{10 \times 3} v^2 = \frac{17}{3} v^2$$

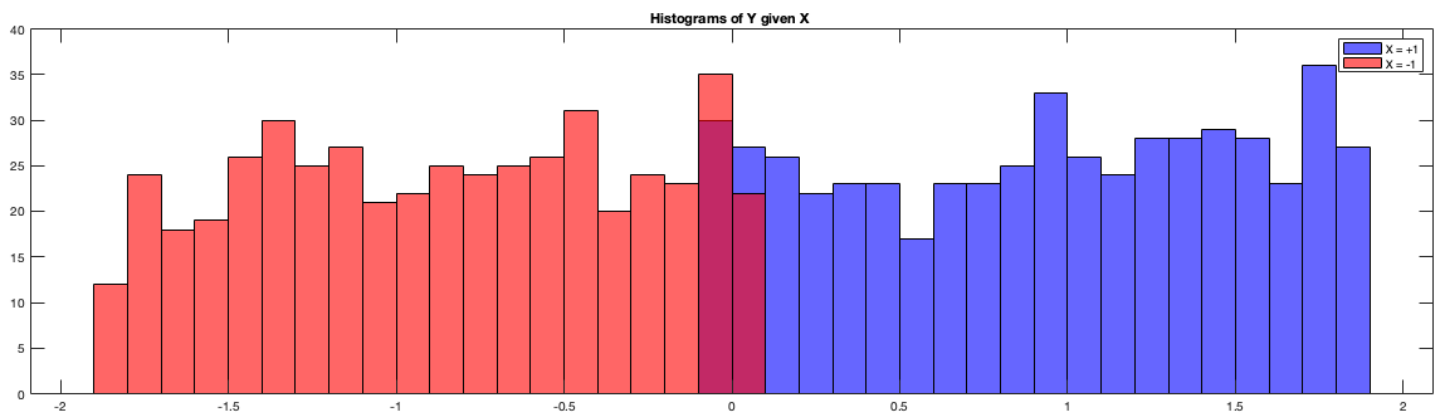
6.a) Noise profiles for $X=1$ & $X=-1$ are different

$$f_{w|X}(w|x=1) = \begin{cases} \frac{1}{2} & , -1.1 \leq w \leq 0.9 \\ 0 & , \text{else} \end{cases} \quad f_{w|X}(w|x=-1) = \begin{cases} \frac{1}{2} & , -0.9 \leq w \leq 1.1 \\ 0 & , \text{else} \end{cases}$$

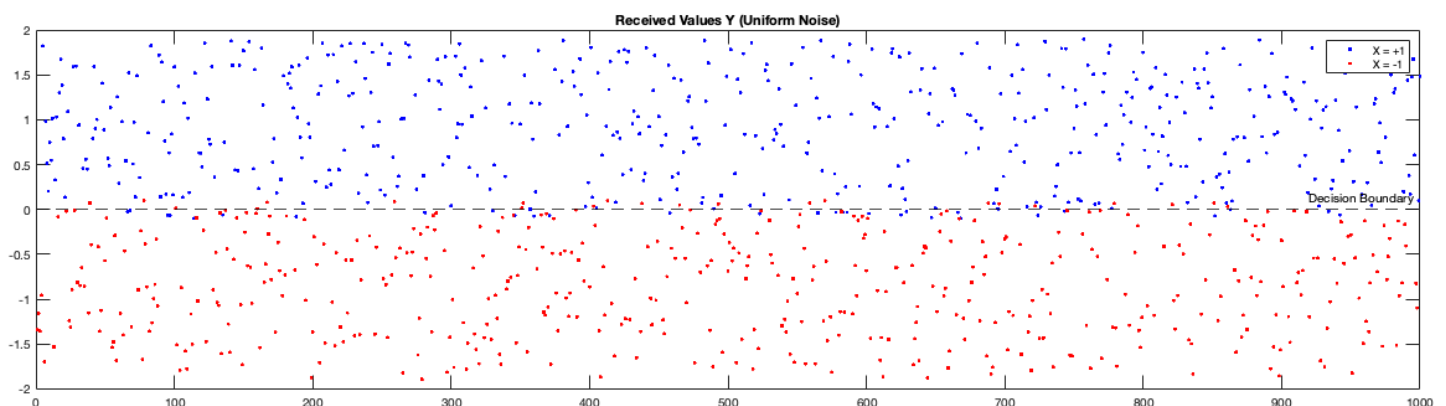


→ 1000 samples is not enough to get a decent uniform distribution.

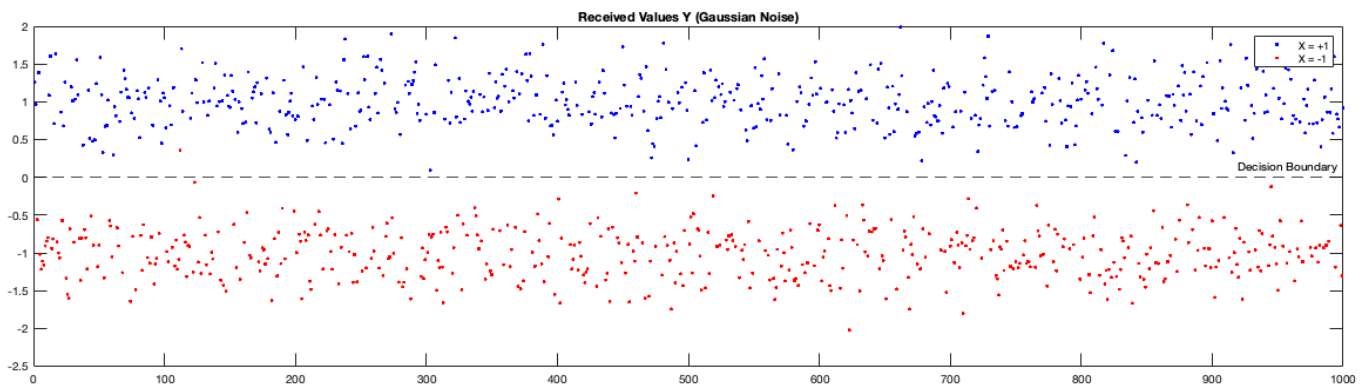
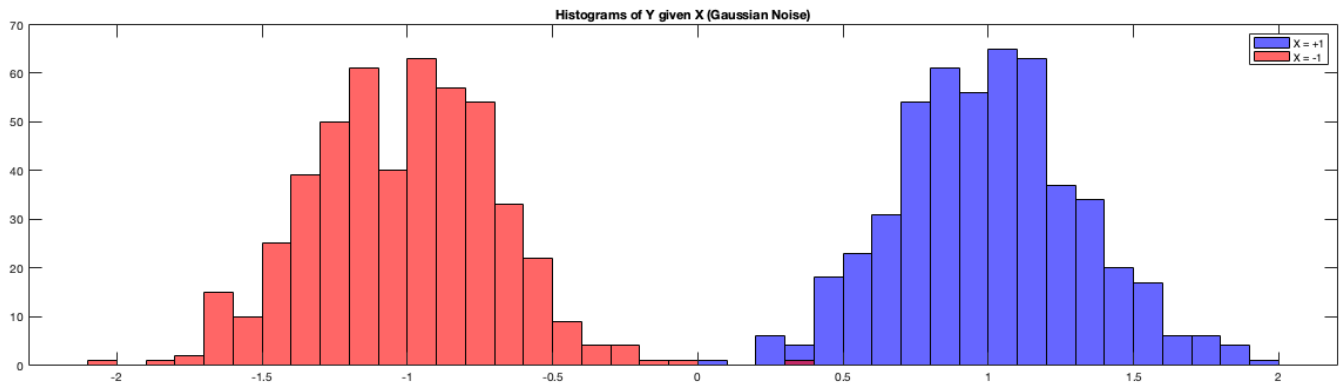
→ In fact when we add symbols X , we get the conditional pdfs in 3(a)



→ We can see below that a small fraction of $X=1$ symbols are below zero and some $X=-1$ are above zero. When I compute the error rate, I get values like 0.046, 0.050, 0.052 which are all around 0.05 in conformity with 3(a)'s theoretical calculation.



6.b) This time noise profile is the same for both symbols and conditional pdf's are more separated compared to previous case. Also noise has zero mean making intersection of pdf's smaller also. Note that below figure is conditional pdf's of Y. To get $f_{Y|X}(Y|X=-1)$ shift red histogram to right by 1, to get $f_{Y|X}(Y|X=1)$ shift blue histogram to the left by 1



- We see that $X=1$ symbols are never below zero and $X=-1$ symbols are rarely above zero, which results in a very small error prob.
- I have usually obtained $P_e = 0$ with some occasional cases of $P_e = 0.001$
- ↳ This is because expected error prob. is really low (7.83×10^{-4}). So we may not always get an error.

6.c)

```
1 N = 1000; % Number of symbols
2 % 1. Generate symbols
3 X = (rand(1000, 1) > 0.5) * 2 - 1;
4
5
6 % 2. Generate noise
7 Y_unif = zeros(1000, 1);
8 W = zeros(1000,1);
9 for i = 1:1000
10     if X(i) == 1
11         W(i) = unifrnd(-1.1,0.9);
12         Y_unif(i) = W(i)+1;
13     else
14         W(i) =unifrnd(-0.9,1.1);
15         Y_unif(i) = W(i)-1;
16     end
17 end
18
19 % Plots results
20 figure; subplot(2,1,1);
21 histogram(Y_unif(X==1), 'FaceColor', 'b', 'BinWidth',0.1); hold on;
22 histogram(Y_unif(X== -1), 'FaceColor', 'r', 'BinWidth',0.1);title('Histograms of Y given X '); legend('X = +1', 'X = -1');
23
24 subplot(2,1,2);
25 plot(find(X==1), Y_unif(X==1), 'b. '); hold on;
26 plot(find(X== -1), Y_unif(X== -1), 'r. ');
27 yline(0, 'k--', 'Decision Boundary');title('Received Values Y (Uniform Noise)'); legend('X = +1', 'X = -1');
28
29 figure;
30 histogram(W(X==1), 'FaceColor', 'b', 'BinWidth',0.1); hold on;
31 histogram(W(X== -1), 'FaceColor', 'r', 'BinWidth',0.1);title('Histograms of W given X'); legend('X = +1', 'X = -1');
32
33 % Error Calculation
34 X_hat_unif = sign(Y_unif);
35 error_unif = sum(X_hat_unif ~= X) / N; fprintf('Uniform Noise Error Probability: %.4f\n', error_unif);
36
37 % Using Gaussian noise (mean 0, variance 0.1)
38 sigma = sqrt(0.1);
39 W_gauss = sigma * randn(N, 1);
40 Y_gauss = X + W_gauss;
41
42 % Plotting
43 figure; subplot(2,1,1);
44 histogram(Y_gauss(X==1), 'FaceColor', 'b'); hold on;
45 histogram(Y_gauss(X== -1), 'FaceColor', 'r');title('Histograms of Y given X (Gaussian Noise)'); legend('X = +1', 'X = -1');
46
47 subplot(2,1,2);
48 plot(find(X==1), Y_gauss(X==1), 'b. '); hold on;
49 plot(find(X== -1), Y_gauss(X== -1), 'r. '); yline(0, 'k--', 'Decision Boundary'); title('Received Values Y (Gaussian Noise)');
50 legend('X = +1', 'X = -1');
51
52 % Error Calculation
53 X_hat_gauss = sign(Y_gauss);
54 error_gauss = sum(X_hat_gauss ~= X) / N; fprintf('Gaussian Noise Error Probability: %.4f\n', error_gauss);
```